

IMAT MOCK EXAM 9: Hard Calibration

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 35/90.0 · Mock exam

Section A - Logical Reasoning & Reading (Q1-9)

1. "Researchers followed two groups of office workers for a year. Group A worked in a building where filtered air was fully exchanged every 20 minutes; Group B worked in an identical building where the air was exchanged every two hours. Temperature, lighting and noise were matched, and neither the workers nor the testers knew which building a participant occupied. Group A scored higher on decision-making tests taken at the end of each working day, with the widest gap on days of high outdoor pollution."

Which statement is most strongly supported by the passage?

- A)** Ventilation rate may influence workers' decision-making, and the difference shows most clearly on days of high outdoor pollution.
- B)** The study proves that outdoor air pollution directly impairs decision-making.
- C)** Temperature, lighting and noise have no effect on decision-making performance.
- D)** The workers in Group A were more motivated than the workers in Group B.
- E)** Indoor air should never be exchanged with outdoor air.

2. "Last year a city began charging drivers to enter its central zone. Within six months, car traffic in the zone had fallen by a quarter and buses ran measurably faster. A survey found that most people entering the zone travelled by public transport, and shopkeepers inside the zone reported slightly lower revenues. The money raised from the charge was used to extend two bus routes."

Which statement is best supported by the passage?

- A)** Buses run faster because drivers have abandoned public transport.
- B)** Car journeys into the central zone fell, but the passage does not establish what caused the fall in shop revenues.
- C)** The charge was introduced mainly in order to raise money for bus routes.
- D)** The fall in shop revenues was caused by the traffic charge.
- E)** Most visitors have stopped coming to the central zone.

3. Every student who passed the practical examination attended all of the laboratory sessions. Some of the students who attended all of the laboratory sessions were awarded a scholarship. No scholarship winner was charged a laboratory fee.

Which one of the following must be true?

- A)** Some students who passed the practical examination were not charged a laboratory fee.
- B)** Some scholarship winners passed the practical examination.
- C)** All students who were charged a laboratory fee failed the practical examination.
- D)** Every student who was not charged a laboratory fee attended all of the laboratory sessions.
- E)** Some students who attended all of the laboratory sessions were not charged a laboratory fee.

4. Exactly one of four office workers jammed the shared printer, and exactly one of the four statements they made about the incident is true. Dana said: Erik jammed the printer. Erik said: Farid did not jam the printer. Farid said: I did not jam the printer. Gus said: Erik did not jam the printer.

Who jammed the printer?

- A)** Dana
- B)** No assignment fits: the statements are inconsistent.
- C)** Erik
- D)** Farid
- E)** Gus

5. A town installed automatic speed cameras along its main road. During the following year, accidents on that road fell by 30 per cent, and the council concluded that the cameras had slowed drivers down and so prevented accidents.

Which one of the following findings would most weaken the council's argument?

- A)** Repairing the cameras cost the council more than it had expected to save.
- B)** The cameras recorded several thousand speeding violations during the year.
- C)** Accidents on other roads in the same town fell by about 30 per cent over the same year, although no cameras were installed there.
- D)** The speed limit on the main road was not changed when the cameras were installed.
- E)** Most drivers questioned said that the cameras made them feel constantly watched.

6. A safety rule states: whenever the fire alarm sounds, all staff evacuate the building.

Which one of the following statements is logically equivalent to this rule?

- A)** If the staff evacuate the building, then the fire alarm has sounded.
- B)** If the staff do not evacuate the building, then the fire alarm has not sounded.
- C)** Either the fire alarm sounds or the staff evacuate the building.
- D)** If the fire alarm does not sound, then the staff do not evacuate the building.
- E)** The staff evacuate the building if and only if the fire alarm sounds.

7. Six friends stand in a row for a photograph, in positions numbered 1 to 6 from left to right. Jana stands in position 1. Femi stands somewhere to the right of Gita. Hugo stands immediately to the left of Kofi. Ivo stands somewhere to the right of Femi.

Which one of the following is NOT possible?

- A)** Gita stands in position 5.
- B)** Ivo stands in position 4.
- C)** Hugo stands in position 2.
- D)** Femi stands in position 3.
- E)** Kofi stands in position 5.

8. The four lectures Anthropology, Botany, Chemistry and Demography are scheduled one per hour at 9:00, 11:00, 13:00 and 15:00. Chemistry is scheduled earlier than Demography. Botany is not scheduled at 9:00. Anthropology is scheduled either immediately before or immediately after Botany.

Which one of the following must be true?

- A)** Botany is scheduled at 13:00.
- B)** Chemistry is scheduled at 15:00.
- C)** Demography is not scheduled at 9:00.
- D)** Anthropology is scheduled at 9:00.
- E)** Chemistry is scheduled at 11:00.

9. Five entries, P, Q, R, S and T, are ranked from 1st (best) to 5th in a short-story competition. T is ranked third. S is ranked better than R. R is ranked better than Q.

Which one of the following must be true?

- A)** P is ranked second.
- B)** R is ranked fourth.
- C)** S is ranked first.
- D)** S is ranked better than P.
- E)** Q is ranked either fourth or fifth.

Section B - Biology (Q10-32)

- 10.** A protein's folded shape is held partly by ionic bonds (salt bridges) between the side chain of a lysine residue and the side chain of an aspartate residue. At pH 7 the lysine side chain is positively charged and the aspartate side chain is negatively charged. The protein is transferred to a solution at pH 2, where the aspartate side chain becomes protonated and loses its charge, while the lysine side chain stays charged. Which outcome is expected?
- A)** The primary structure is shortened, because the bond between the two side chains is one kind of peptide bond.
 - B)** The salt bridge is unaffected, but every peptide bond in the backbone breaks at pH 2.
 - C)** The salt bridge becomes stronger, because two uncharged groups attract each other more strongly than two charged groups.
 - D)** The salt bridge breaks and this part of the tertiary structure becomes less stable, while the amino acid sequence is unchanged.
 - E)** A new salt bridge forms immediately, because at pH 2 the aspartate side chain becomes positively charged.
- 11.** A patient takes a drug that keeps the pH of the stomach contents close to 7 instead of the usual strongly acidic value, but leaves the small intestine unchanged. Pepsin works optimally near pH 2, while trypsin works optimally near pH 8. Which outcome is expected?
- A)** Protein digestion continues unchanged, because enzymes adjust their pH optimum to their surroundings.
 - B)** Both enzymes lose almost all activity, because digestive enzymes work only in strongly acidic conditions.
 - C)** Pepsin activity in the stomach falls sharply, while trypsin activity in the small intestine is essentially unaffected.
 - D)** Pepsin activity in the stomach rises, because pepsin neutralises the effect of the drug.
 - E)** Trypsin activity in the small intestine falls to zero, because trypsin requires the acidity produced by the stomach.
- 12.** Glucose units released from glycogen inside a muscle cell enter glycolysis as glucose 6-phosphate, so the hexokinase step, which consumes 1 ATP, is bypassed, while the other ATP-consuming step of glycolysis still occurs. The muscle ferments these units entirely to lactate. What is the net ATP yield per glucose unit?
- A) 0 B) 2 C) 3 D) 4 E) 6**
- 13.** In the epithelial cells lining the small intestine, glucose is absorbed from the lumen by a transporter that carries glucose together with sodium ions, driven by the sodium concentration gradient. That gradient is maintained by the sodium-potassium pump, which hydrolyses ATP. A poison inactivates the sodium-potassium pump of these cells. Which outcome is expected?
- A)** The cells begin to secrete glucose back into the lumen, because the pump normally carries glucose inward.
 - B)** The sodium-potassium pump switches to transporting glucose, so absorption continues unchanged.
 - C)** Only the movement of potassium is affected, because sodium and potassium cross the membrane through separate systems.
 - D)** The sodium gradient across the membrane decays, so the sodium-coupled glucose transporter stops working and glucose absorption by this route fails.
 - E)** Glucose absorption by the sodium-coupled transporter continues at a normal rate, because this transporter does not use ATP directly.

- 14.** A plant leaf cell is placed in a solution whose concentration of non-penetrating solutes is much higher than that of the cell sap. Which outcome is expected?
- A)** Water enters the cell until the cell bursts, because plant cells have no way of stopping water uptake.
 - B)** Water leaves the cell, the protoplast pulls away from the cell wall, and the cell does not burst.
 - C)** Water enters the cell and builds turgor pressure against the wall.
 - D)** There is no net movement of water, because the cell wall blocks osmosis.
 - E)** Water leaves the cell until the cell bursts, because the wall cannot resist shrinking.
- 15.** A drug stops the transcription of ribosomal RNA in the nucleolus of a eukaryotic cell, while the transcription of protein-coding genes and the rest of the cell's metabolism continue intact for several hours. Which outcome is expected?
- A)** Messenger RNA accumulates but is translated less efficiently, because no new ribosomal subunits are assembled, so the rate of protein synthesis falls.
 - B)** Messenger RNA is destroyed as soon as it leaves the nucleus, because the nucleolus protects it.
 - C)** DNA replication speeds up, because the freed RNA polymerase begins copying DNA.
 - D)** Only the synthesis of proteins destined for secretion stops, because those proteins are made by the nucleolus itself.
 - E)** Protein synthesis continues unchanged, because ribosomes are assembled in the cytoplasm without the nucleolus.
- 16.** In a eukaryotic gene, a single base substitution occurs well upstream of the coding sequence, within the promoter region where RNA polymerase binds, and the change weakens this binding. The coding sequence itself is unchanged. Which outcome is expected?
- A)** The reading frame of the gene shifts, so the protein differs completely downstream of the mutation.
 - B)** The gene can never be transcribed again, because a damaged promoter destroys the DNA.
 - C)** The cell is unaffected in every way, because promoter changes cannot influence protein production.
 - D)** One amino acid in the protein is replaced by a different one, because every base substitution changes the protein.
 - E)** The amino acid sequence of the protein is unchanged, but the protein is produced in reduced amounts, because transcription of the gene is less efficient.
- 17.** During meiosis I in a human oocyte, one pair of homologous chromosomes fails to separate (nondisjunction). Meiosis II then proceeds normally, and the resulting egg cell is fertilised by a normal sperm. Which statement is correct?
- A)** The zygote always has 47 chromosomes, because nondisjunction always produces extra chromosomes.
 - B)** The egg is always inviable, because cells with unpaired homologous chromosomes cannot be fertilised.
 - C)** The four products have 24, 24, 23 and 23 chromosomes, because only the second division is disturbed.
 - D)** Two of the four products are normal, and fertilisation produces a normal zygote in half of the cases.
 - E)** All four products of meiosis have an abnormal chromosome number, and fertilisation by a normal sperm produces a zygote with either 45 or 47 chromosomes.
- 18.** Two guinea pigs, both heterozygous for coat colour (Bb) and for hair length (Ll), are crossed. The two genes assort independently. What is the probability that an offspring shows both recessive traits, that is, brown coat (bb) and long hair (ll)?
- A)** 1/16 **B)** 1/8 **C)** 1/4 **D)** 3/16 **E)** 9/16
- 19.** In a population in Hardy-Weinberg equilibrium, one child in 2,500 is born with a recessive genetic disease. Assuming random mating and a large population, roughly what fraction of the population are carriers (heterozygotes) for the disease allele?
- A)** 1/2500 **B)** 1/50 **C)** 1/25 **D)** 2/25 **E)** 49/50

- 20.** Colchicine prevents the assembly of spindle microtubules. A cell that has completed DNA replication is exposed to colchicine as it enters mitosis. At which stage, and with which appearance, will treated cells accumulate?
- A)** In anaphase, because colchicine triggers the separation of the chromatids.
 - B)** In interphase, because the cell returns to G1 without dividing.
 - C)** In prophase, because spindle formation is required for chromosome condensation.
 - D)** At metaphase, with fully condensed chromosomes each consisting of two sister chromatids and no separation of the chromatids.
 - E)** In telophase, because the nuclear envelope reforms without spindle function.
- 21.** A toxin blocks the proton channel of mitochondrial ATP synthase, so protons can no longer flow back into the matrix, while the electron transport chain still transfers electrons to oxygen at first. In cells whose mitochondria are treated with this toxin, which statement is correct?
- A)** The cells switch to photosynthesis in order to regenerate their NADPH.
 - B)** Electron transport soon stops because the growing proton gradient opposes further pumping, NADH accumulates and is no longer reoxidised, and glycolysis can only continue by converting pyruvate to lactate.
 - C)** Glycolysis continues normally, because glycolysis never needs NAD⁺ at any step.
 - D)** The proton gradient dissipates through the blocked ATP synthase and electron transport accelerates above normal.
 - E)** Electron transport continues at its normal rate, because oxygen is still available and ATP synthesis is a separate process.
- 22.** After death, calcium ions leak out of the sarcoplasmic reticulum and are never pumped back, and ATP production has stopped. Cross-bridges form between myosin heads and actin. Which statement explains why the muscles then become rigid (rigor mortis)?
- A)** The actin filaments shorten permanently, pulling the muscle rigid.
 - B)** Without ATP the myosin heads cannot bind actin at all, so the fibres separate and harden.
 - C)** The lack of ATP causes the sarcoplasmic reticulum to release more calcium, which stiffens the muscle membrane.
 - D)** Tropomyosin detaches from actin because its position depends on ATP rather than on calcium.
 - E)** Myosin heads remain locked onto actin because ATP is needed to detach them, and with no ATP the cross-bridges cannot break, so the muscle stays contracted.
- 23.** The aortic semilunar valve of a patient is narrowed (stenosed), so the opening through which blood leaves the left ventricle is reduced. Which chain of consequences follows?
- A)** Blood flows back from the aorta into the left ventricle during diastole, and the right ventricle enlarges first.
 - B)** The left ventricle must generate a higher pressure to eject blood, ejection takes longer, and the ventricle wall thickens over time under the extra load.
 - C)** The left atrium enlarges because it pumps blood directly into the aorta.
 - D)** The mitral valve is held open during systole, which raises the pressure in the left atrium.
 - E)** Blood pressure in the pulmonary arteries falls, because the lungs receive less blood.
- 24.** In a healthy chest, the pressure in the pleural cavity between the lung and the chest wall is lower than atmospheric, which keeps the lung expanded against the chest wall. A fractured rib tears the lung surface and lets air leak into the right pleural cavity until the pressure there equals atmospheric pressure. What happens to the right lung, and why?
- A)** The right lung collapses inward, because its elastic recoil is no longer opposed by the sub-atmospheric pleural pressure, while the chest wall springs outward.
 - B)** The lung and the chest wall are drawn toward each other with extra force, keeping the lung inflated.
 - C)** The right lung inflates further, because the incoming air pushes directly into the lung tissue.
 - D)** The right lung collapses because the air destroys the elastic fibres of the alveolar walls.
 - E)** The right lung is unaffected, because the diaphragm alone keeps it inflated.

25. Carbon monoxide binds to haemoglobin at the same sites as oxygen, with an affinity more than 200 times greater, and it is released only slowly. A fire victim is breathing air containing carbon monoxide. A pulse oximeter cannot distinguish carboxyhaemoglobin from oxyhaemoglobin. Which statement is correct?

- A)** Tissue oxygen delivery rises, because carbon monoxide increases the affinity of haemoglobin for oxygen.
- B)** The blood carries less oxygen, and a pulse oximeter immediately shows a falsely low reading, which makes diagnosis simple.
- C)** Oxygen transport is normal, because carbon monoxide displaces carbon dioxide from haemoglobin rather than oxygen.
- D)** The blood carries far less oxygen than its haemoglobin capacity allows, yet a pulse oximeter still reads a high saturation because carbon monoxide bound to haemoglobin looks like oxygen-bound haemoglobin.
- E)** Carbon monoxide only reduces the small amount of oxygen dissolved in plasma, leaving haemoglobin transport untouched.

26. A glucose molecule is absorbed from the lumen of the small intestine into a capillary of an intestinal villus. Which of the following gives the correct route it takes on the way to the hepatic vein?

- A)** Intestinal villus capillary, hepatic artery, liver sinusoids, hepatic portal vein
- B)** Intestinal villus capillary, hepatic vein, liver sinusoids, hepatic portal vein
- C)** Intestinal villus capillary, hepatic portal vein, liver sinusoids, hepatic vein
- D)** Intestinal villus capillary, vena cava, right atrium, hepatic portal vein
- E)** Intestinal villus capillary, lacteal, thoracic duct, hepatic vein

27. A person takes tablets containing thyroxine (T4) every day in fairly high doses, without medical supervision. The normal feedback control of the thyroid axis still works correctly. Which pattern of blood hormone levels is expected?

- A)** TSH low and T4 high, because the extra T4 suppresses the hypothalamus and the pituitary through normal negative feedback.
- B)** TSH and T4 both normal, because thyroxine taken by mouth is destroyed before reaching the blood.
- C)** TSH high and T4 high, because the pituitary reacts to high T4 by secreting more TSH.
- D)** TRH and TSH both high, because the thyroid gland cannot respond to the tablets.
- E)** TSH high and T4 low, because the tablets stop the thyroid gland from working.

28. A hill walker walks for hours in extreme heat, loses a large volume of sweat and drinks almost nothing. Plasma volume falls and the plasma becomes more concentrated. Which set of responses is expected?

- A)** Aldosterone secretion falls, so more sodium is lost in the urine.
- B)** The glomerular filtration rate rises sharply and urine volume increases.
- C)** ADH secretion stays unchanged, because ADH responds only to blood pressure and never to plasma concentration.
- D)** ADH secretion falls and the kidneys produce large volumes of dilute urine.
- E)** ADH secretion rises, the collecting ducts reabsorb more water so the urine becomes concentrated, and thirst is stimulated.

29. A person with type 1 diabetes injects insulin but then skips the next meal and takes vigorous exercise, and the blood glucose level begins to fall below normal. Which sequence of responses follows in this person?

- A)** The pancreas secretes additional insulin, which corrects the glucose level.
- B)** Glucagon and adrenaline secretion rise, the liver breaks down glycogen and releases glucose into the blood, and the person feels warning symptoms such as hunger and trembling.
- C)** The liver converts glucose into glycogen faster, storing away the effect of the injected insulin.
- D)** The kidneys stop filtering glucose, which keeps the blood level constant.
- E)** Glucagon secretion falls, so the muscles take up more glucose from the blood.

- 30.** The oesophagus is lined by stratified squamous epithelium many cells thick, while the small intestine is lined by a single layer of columnar cells. Which statement best explains this difference?
- A)** The two linings do the same job, and the difference is a random outcome of embryonic development.
 - B)** A single layer of columnar cells is stronger, which suits the intestine as the first part of the gut to receive food.
 - C)** Stratified squamous epithelium secretes more digestive enzymes, which is why the oesophagus digests food first.
 - D)** The oesophagus needs a many-layered, abrasion-resistant lining against swallowed food, while the intestine needs a thin single layer specialised for absorption.
 - E)** The oesophagus absorbs nutrients more slowly than the intestine and needs extra layers to compensate.
- 31.** A patient tears the meniscus, a pad of cartilage in the knee, and is told that it heals much more slowly than a similar tear in the skin. Which feature of cartilage best explains this?
- A)** Cartilage is buried under skin everywhere in the body, so repair cells cannot physically reach it.
 - B)** The cartilage matrix is made of keratin, which never grows back after damage.
 - C)** Cartilage cells divide faster than skin cells but build a weaker matrix.
 - D)** Cartilage contains few or no blood vessels, so repair cells and their building materials reach the injury only slowly by diffusion.
 - E)** Cartilage is a type of nervous tissue, so repair signals cannot reach the injury.
- 32.** A large beetle population has an allele frequency of 0.4 for allele A at one locus. A flood leaves only ten survivors: six AA, three Aa and one aa, and these found a new population. What is the frequency of allele A among the survivors, and what does the change from 0.4 illustrate?
- A)** 0.15, and the change illustrates gene flow from another population.
 - B)** 0.40, and the change illustrates the constancy required by the Hardy-Weinberg principle.
 - C)** 0.60, and the change illustrates natural selection favouring allele A.
 - D)** 0.75, and the change illustrates genetic drift caused by a population bottleneck.
 - E)** 0.80, and the change illustrates a new mutation increasing allele A.

Section C - Chemistry (Q33-47)

- 33.** Consider three pairs of organic compounds. Pair P: butan-1-ol and butan-2-ol. Pair Q: (2R)-butan-2-ol and (2S)-butan-2-ol. Pair R: cis-pent-2-ene and trans-pent-2-ene. Which of the following statements is correct?
- A)** P and Q are pairs of enantiomers, while R is a pair of constitutional isomers.
 - B)** All three pairs are constitutional isomers, since each pair shares the same molecular formula.
 - C)** R is a pair of enantiomers, because the cis and trans forms are mirror images of each other.
 - D)** P is a pair of stereoisomers, since the two alcohols differ only in the position of the OH group.
 - E)** Q and R are pairs of stereoisomers, while P is a pair of constitutional isomers.
- 34.** 4.8 g of methane ($M = 16 \text{ g/mol}$) is mixed with 4.8 g of oxygen gas ($M = 32 \text{ g/mol}$) and ignited, and the reaction $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$ proceeds until one reactant is used up. Which statement is correct? (Molar volume of any gas at STP = $22.4 \text{ dm}^3/\text{mol}$)
- A)** 1.68 dm^3 of CO_2 (measured at STP) forms and 3.6 g of methane remains.
 - B)** 1.68 dm^3 of CO_2 (measured at STP) forms and no methane remains.
 - C)** 1.68 dm^3 of CO_2 (measured at STP) forms and 1.8 g of methane remains.
 - D)** 3.36 dm^3 of CO_2 (measured at STP) forms and the oxygen is in excess.
 - E)** 3.36 dm^3 of CO_2 (measured at STP) forms and 1.8 g of methane remains.
- 35.** 0.40 g of sodium hydroxide ($M = 40 \text{ g/mol}$), a strong base that is fully dissociated in water, is dissolved and made up to exactly 1.0 dm^3 of solution at 25°C . Then 10 cm^3 of this solution is diluted with water to a final volume of 1.0 dm^3 . What is the pH of the final solution? ($K_w = 1.0 \times 10^{-14}$ at 25°C)
- A) 8 B) 9 C) 10 D) 11 E) 12**

- 36.** At 25 °C the ionic product of water is $K_w = 1.0 \times 10^{-14}$, but at a higher temperature K_w is larger: in one such case $K_w = 1.0 \times 10^{-13}$. What is the pH of a 0.010 mol/dm³ solution of sodium hydroxide, fully dissociated, at this higher temperature?
A) 2 B) 6.5 C) 11 D) 12 E) 13
- 37.** 20 cm³ of 0.010 mol/dm³ hydrochloric acid, a strong acid that is fully ionised, is diluted with water to a final volume of 400 cm³. Taking $\log_{10} 2 = 0.30$, what is the pH of the diluted solution?
A) 2.30 B) 3.00 C) 3.10 D) 3.30 E) 4.00
- 38.** Four pure molecular liquids with similar molar masses are: propane CH₃CH₂CH₃ (Mr 44), propanone CH₃COCH₃ (Mr 58), ethanol C₂H₅OH (Mr 46) and ethanoic acid CH₃COOH (Mr 60). In the liquid state, two ethanoic acid molecules are held together by two hydrogen bonds each, and propanone has no O-H or N-H bond. Which option places them in order of increasing boiling point?
A) ethanoic acid < ethanol < propanone < propane
B) propane < propanone < ethanol < ethanoic acid
C) propane < propanone < ethanoic acid < ethanol
D) propanone < propane < ethanol < ethanoic acid
E) propane < ethanol < propanone < ethanoic acid
- 39.** For the reaction $A + 2B \rightarrow \text{products}$, experiments at fixed temperature show that the rate is proportional to the concentration of A and to the square of the concentration of B. If the concentration of A is doubled and the concentration of B is tripled, by what factor does the reaction rate increase?
A) 6 B) 8 C) 9 D) 12 E) 18
- 40.** What is the average oxidation number of a sulfur atom in the tetrathionate ion, S₄O₆²⁻, if the four sulfur atoms are treated as equivalent?
A) +2 B) +2.5 C) +3 D) +5 E) +10
- 41.** In which species does nitrogen have the same oxidation number as sulfur has in the sulfite ion, SO₃²⁻?
A) NO₂ B) NH₃ C) NO D) N₂ E) N₂O
- 42.** A substance decomposes by first-order kinetics with a half-life of 20 minutes at 25 °C. A sealed 2.0 dm³ vessel initially contains 0.80 mol of the substance. What is its concentration after 60 minutes?
A) 0.025 mol/dm³ B) 0.040 mol/dm³ C) 0.050 mol/dm³ D) 0.10 mol/dm³ E) 0.20 mol/dm³
- 43.** Sample X contains 4.4 g of carbon dioxide, CO₂, and sample Y contains 4.9 g of sulfuric acid, H₂SO₄ (M = 98 g/mol). (C = 12, O = 16; Avogadro constant = 6.02×10^{23} per mol.) Which statement about the oxygen atoms in the two samples is correct?
A) Each sample contains 1.2×10^{23} oxygen atoms.
B) Each sample contains 2.4×10^{23} oxygen atoms.
C) Sample X contains 2.4×10^{23} oxygen atoms and sample Y contains 1.2×10^{23} .
D) Each sample contains 6.0×10^{22} oxygen atoms.
E) Sample X contains 1.2×10^{23} oxygen atoms and sample Y contains 2.4×10^{23} .
- 44.** 50 cm³ of 0.20 mol/dm³ hydrochloric acid, fully ionised, is mixed with 150 cm³ of 0.10 mol/dm³ sodium hydroxide, fully dissociated. What is the pH of the resulting solution at 25 °C? ($K_w = 1.0 \times 10^{-14}$; $\log 2.5 = 0.40$)
A) 1.60 B) 11.60 C) 12.00 D) 12.40 E) 13.00
- 45.** 1.08 g of aluminium (M = 27 g/mol) is added to 100 cm³ of 0.10 mol/dm³ sulfuric acid and reacts completely as far as possible: $2\text{Al} + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2$. Which statement is correct? (Molar volume of a gas at STP = 22.4 dm³/mol)
A) 0.224 dm³ of H₂ (measured at STP) is released and the acid is in excess.
B) 0.224 dm³ of H₂ (measured at STP) is released and aluminium is in excess.
C) 0.224 dm³ of H₂ (measured at STP) is released and no aluminium remains.
D) 0.448 dm³ of H₂ (measured at STP) is released and aluminium is in excess.
E) 1.344 dm³ of H₂ (measured at STP) is released and the acid is in excess.

46. A student evaporates water from 500 cm³ of a 0.20 mol/dm³ sodium chloride solution until the volume is exactly 200 cm³, with no loss of solute. The concentration of the final solution is:

- A)** 0.08 mol/dm³ **B)** 0.10 mol/dm³ **C)** 0.20 mol/dm³ **D)** 0.40 mol/dm³ **E)** 0.50 mol/dm³

47. Two solutions at 25 °C have pH values of 3.5 and 5.5. The hydrogen ion concentration of the solution with pH 3.5 is how many times that of the solution with pH 5.5?

- A)** 2 **B)** 20 **C)** 100 **D)** 1000 **E)** 10000

Section D - Mathematics (Q48-54)

48. Given that $\log_{10} 5 = 0.70$ and $\log_{10} 3 = 0.48$, what is the value of $\log_{10} 45$?

- A)** 1.18 **B)** 1.30 **C)** 1.66 **D)** 1.88 **E)** 2.18

49. For which values of k does the equation $x^2 + kx + 9 = 0$ have no real solutions?

- A)** $k \leq -6$ or $k \geq 6$
B) $-6 \leq k \leq 6$
C) all real values of k
D) $k < -6$ or $k > 6$
E) $-6 < k < 6$

50. A solid metal sphere of radius 3 cm is melted down and recast into a solid right circular cylinder of radius 2 cm. What is the height of the cylinder? (Volume of a sphere = $\frac{4}{3}\pi r^3$; volume of a cylinder = $\pi r^2 h$)

- A)** 3 cm **B)** 6 cm **C)** 9 cm **D)** 18 cm **E)** 36 cm

51. A bag contains 3 red balls and 5 blue balls. Two balls are drawn at random without replacement. What is the probability that the two balls are the same colour?

- A)** $\frac{3}{28}$ **B)** $\frac{5}{14}$ **C)** $\frac{13}{28}$ **D)** $\frac{15}{56}$ **E)** $\frac{15}{28}$

52. A toy train runs along a straight track away from its start point, and its distance from the start is recorded at one-minute intervals: 0 m at 0 min, 120 m at 1 min, 300 m at 2 min and 540 m at 3 min. What is the greatest average speed over any one-minute interval of the journey, expressed in m/s?

- A)** 2.5 m/s **B)** 3.0 m/s **C)** 4.0 m/s **D)** 5.0 m/s **E)** 9.0 m/s

53. What is the value of $\sqrt{50} + \sqrt{18} - \sqrt{8}$, written in the form $a\sqrt{2}$?

- A)** 2 **B)** 4 **C)** 6 **D)** 8 **E)** 10

54. For two events A and B, $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \text{ and } B) = 0.2$. What is $P(A \text{ or } B)$?

- A)** 0.2 **B)** 0.5 **C)** 0.6 **D)** 0.7 **E)** 0.9

Section E - Physics (Q55-60)

55. A trolley moves along a straight track and its velocity is recorded every 2.0 s: 0 m/s at 0 s, 10 m/s at 2.0 s, 14 m/s at 4.0 s, 14 m/s at 6.0 s and 6 m/s at 8.0 s. What is the magnitude of the trolley's acceleration between 6.0 s and 8.0 s?

- A)** 2 m/s² **B)** 4 m/s² **C)** 5 m/s² **D)** 6 m/s² **E)** 8 m/s²

56. A car travels around a circular bend of radius 50 m at a constant speed of 10 m/s. Which of the following statements is correct?

- A)** The acceleration is 2.0 m/s², directed towards the centre of the bend.
B) The acceleration is 2.0 m/s², directed along the direction of travel.
C) The acceleration is 2.0 m/s², directed away from the centre of the bend.
D) The acceleration is zero, because the speed is constant.
E) The acceleration is 0.20 m/s², directed towards the centre of the bend.

57. A battery of negligible internal resistance supplies 24 V to an 8.0 Ω resistor connected in series with a parallel combination of a 12 Ω resistor and a 6.0 Ω resistor. What current does the battery supply?

- A)** 1.5 A **B)** 2.0 A **C)** 2.4 A **D)** 3.0 A **E)** 6.0 A

58. In an insulated container, 0.40 kg of water at 20 °C is mixed with 0.20 kg of water at 65 °C. Assuming no heat is exchanged with the surroundings, what is the final temperature of the mixture?

- A)** 20 °C **B)** 35 °C **C)** 42.5 °C **D)** 50 °C **E)** 65 °C

59. A farm tank contains a liquid of density 800 kg/m³ filled to a depth of 3.0 m and open to the atmosphere (100 kPa). What is the gauge pressure exerted by the liquid at the bottom of the tank? ($g = 10 \text{ m/s}^2$)

- A)** 2.4 kPa **B)** 8.0 kPa **C)** 24 kPa **D)** 124 kPa **E)** 240 kPa

60. An electric kettle rated at 2.0 kW takes 3.0 minutes to bring its water to the boil. How much electrical energy does the kettle transfer in that time?

- A)** $3.6 \times 10^4 \text{ J}$ **B)** $3.6 \times 10^5 \text{ J}$ **C)** $3.6 \times 10^6 \text{ J}$ **D)** $6.0 \times 10^3 \text{ J}$ **E)** $6.0 \times 10^5 \text{ J}$

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	A	2	B	3	E	4	D	5	C
6	B	7	A	8	C	9	E	10	D
11	C	12	C	13	D	14	B	15	A
16	E	17	E	18	A	19	C	20	D
21	B	22	E	23	B	24	A	25	D
26	C	27	A	28	E	29	B	30	D
31	D	32	D	33	E	34	A	35	C
36	C	37	D	38	B	39	E	40	B
41	A	42	C	43	A	44	D	45	B
46	E	47	C	48	C	49	E	50	C
51	C	52	C	53	C	54	D	55	B
56	A	57	B	58	B	59	C	60	B